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Participants were asked to solve five puzzles designed to test their ability to apply new information, reason inductively and deductively, and use logic. Participants were also asked to complete a 20-item vocabulary and grammar test. The older adults performed significantly worse than the younger adults on the puzzles, but there was no significant difference between the groups on the vocabulary/grammar test.

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When asked to read a short story and then answer a series of questions assessing comprehension (focused condition), older adults performed equally well as the younger adults. However, when asked to read a similar story while wearing headphones playing another story that was to be ignored (distraction condition), older adults performed worse than younger adults when answering questions assessing comprehension of the story they read.

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When asked to memorize 30 neutral items (eg, a key, a pencil, a book), the images of which were presented serially, younger adults performed significantly better when asked to list the items immediately after seeing them and when asked to list the items after performing a *distraction task*. In a second trial, both groups were serially presented with 30 pictures of faces showing positive, negative, or neutral expressions. The groups were later asked to identify these faces among a set of 100. Results are shown in Figure 1.

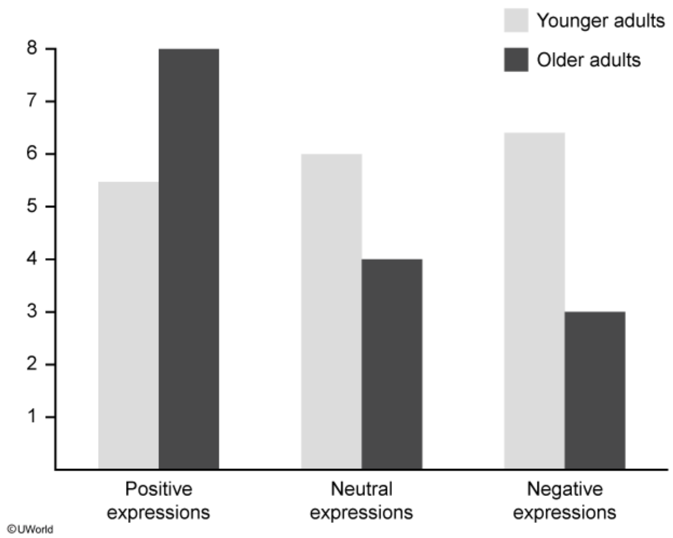

Figure 1 Average number of correctly identified faces for older and younger adult groups

The *cognition studies* and *focus studies* suggest that aging is correlated with declines in:

- A. crystallized intelligence and divided attention. [4%]
- B. crystallized intelligence and selective attention. [5%]
- C. fluid intelligence and divided attention. [40%]
- D. fluid intelligence and selective attention. [48%]

Correct
49% answered correctly

Explanation:

Intelligence	
Fluid	**Crystallized**
Ability to apply *logic & creative thinking* to new situations	Ability to apply *facts & acquired knowledge* to situations

Attention	
Selective	**Divided**
Ability to focus on *one stimulus* or task despite distractions	Ability to attend to *two or more* tasks or stimuli simultaneously

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The cognition and focus studies examined alterations in intelligence and attention in individuals of advanced age. **Intelligence** is the ability to **learn** and **apply new information** and skills, **adapt** to the environment, and **reason** through complex situations. **Attention** is defined as the cognitive processes that **filter** some sensory inputs in order to **focus** on others. Both intelligence and attention can be further subdivided into types:

- Types of intelligence: **Fluid intelligence** is the ability to use logic and creativity to solve novel problems and identify patterns in new situations. **Crystallized intelligence** is the ability to apply skills and knowledge that have already been learned. Fluid intelligence is thought to peak in early adulthood and decline with age, whereas crystallized intelligence is thought to increase with age.
- Types of attention: **Selective attention** is the ability to focus on one stream of information while ignoring other stimuli. **Divided attention** is the ability to attend to more than one stimulus or task at a time.

In the cognition studies, older adults performed poorly compared with younger adults on novel puzzles, suggesting a decline in fluid intelligence. During the focus studies, participants were asked to focus on reading a printed story while ignoring the story on their headphones (distraction condition), which is an example of a selective attention task, on which the older adults performed worse than the younger adults.

(Choices A and B) In the cognitive studies, knowledge of vocabulary and grammar was used as a measure of crystallized intelligence. Older adults performed just as well as the younger adults on this task.

(Choice C) The focus studies did not include any divided attention tasks. In general, research shows that subjects of all ages perform poorly on divided attention tasks.

Educational objective:
Intelligence is the ability to learn, apply, adapt, and reason; fluid intelligence involves logic and creativity whereas crystallized intelligence involves knowledge and skills. Attention is the ability to filter certain stimuli and focus on others; selective attention is focusing on one stimulus or task and divided attention is focusing on more than one.

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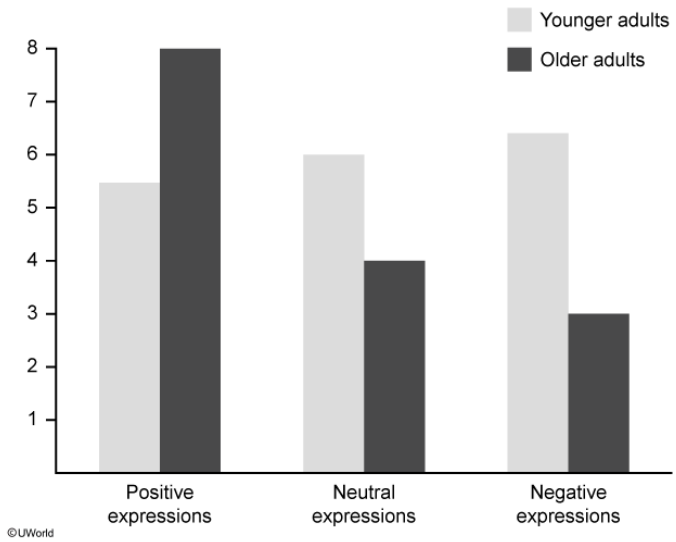

Figure 1 Average number of correctly identified faces for older and younger adult groups

Which conclusion is best supported by the results of the *memory studies*?

- A. Overall, delayed recall of faces declines with age. [9%]
- B. Cued recall of common objects declines with age. [35%]
- C. Recognition of common objects declines with age. [7%]
- D. Recognition of certain faces improves with age. [47%]

Correct
47% answered correctly

Explanation:

Memory retrieval		
Recall Retrieval of information from memory *(eg, fill-in-the-blank tests)*	**Recognition** Identification of previously learned information/target *(eg, multiple-choice tests)*	**Relearning** Re-encoding of information learned but forgotten *(eg, relearning algebra as an adult)*

Memory involves three steps: encoding, storage, and retrieval. **Retrieval** refers to *accessing* the encoded information from storage. There are three types of memory retrieval processes: recall, recognition, and relearning.

- **Recall** is the **retrieval of information** previously encoded. *Free recall* involves remembering without a hint or prompt, whereas *cued recall* involves remembering with the help of a hint or prompt. *Immediate recall* involves the retrieval of information immediately after it was learned, whereas *delayed recall* involves retrieval of information after some time has passed.
- **Recognition** involves the **correct identification** of information that one has been exposed to.
- **Relearning** involves **re-encoding information** that was previously learned but forgotten. Typically, relearning happens much faster than learning something for the first time.

In the memory studies, participants were tested on their abilities to *recall* common objects and to *recognize* a group of faces. According to Figure 1, older adults demonstrated better recognition of faces with positive expressions than younger adults, suggesting improved recognition of certain faces.

(Choice A) The results in Figure 1 suggest an overall age-related decline in *recognition* of faces but improved recognition of faces showing positive expressions. *Delayed recall* of faces was not assessed in this study.

(Choices B and C) The first memory study showed that *free recall* of common objects declines with age. This study did not assess cued recall or recognition of common objects.

Educational objective:
Memory retrieval involves recall, recognition, and relearning. Recall is the retrieval of previously learned information without a prompt (free recall) or with a prompt (cued recall) and can be subdivided into immediate recall and delayed recall. Recognition involves identifying information that one has already been exposed to. Relearning involves re-encoding information that was previously learned but forgotten.

Time spent:0 sec	**QID**:400379
Subject:	**System:**
Behavioral Sciences	Learning, Memory, and Cognition

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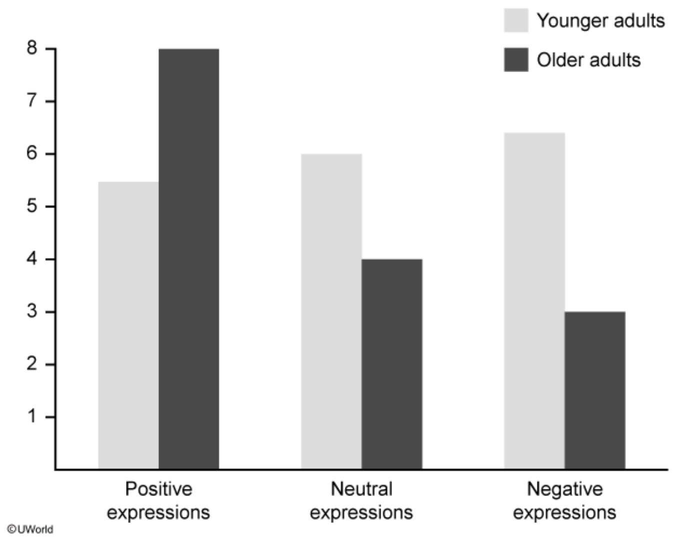

Figure 1 Average number of correctly identified faces for older and younger adult groups

Which of the following conflicts corresponds to the stage of psychosocial development most appropriate for the age of the older adult study participants?

- A. Autonomy vs. shame [4%]
- B. Industry vs. inferiority [3%]
- C. Integrity vs. despair [85%]
- D. Intimacy vs. isolation [6%]

Correct
86% answered correctly

Explanation:

Erikson's stages of psychosocial development			
Stage	**~Age (years)**	**Conflict**	**Successful resolution**
1. Infancy	0–1	Trust vs. mistrust	Ability to have faith in others
2. Early childhood	1–3	Autonomy vs. shame/doubt	Sense of self-control & independence
3. Play age	3–6	Initiative vs. guilt	Ability to take initiative with peers

4. School age	6–12	Industry vs. inferiority	Sense of confidence in skills & abilities
5. Adolescence	12–20	Identity vs. confusion	Sense of self-identity
6. Early adulthood	20–40	Intimacy vs. isolation	Ability to commit to & love others
7. Middle age	40–65	Generativity vs. stagnation	Concern for others & society
8. Old age	>65	Integrity vs. despair	Sense of accomplishment & fulfillment

Erik **Erikson's psychosocial theory** of personality development includes eight stages that represent the human life span. Each stage is associated with an age-related **crisis**, which is an opportunity for individual growth and social development. Resolution does not necessarily occur in each stage before an individual moves onto the next (unresolved conflict forms the basis for adult psychopathology and other maladaptive behaviors).

- Trust vs. mistrust: Infants (0–1 years) with sensitive and attentive caregivers develop a sense of trust; those with inconsistent care do not.
- Autonomy vs. shame/doubt: Toddlers (1–3 years) who are encouraged will develop independence; those who are scolded for failure will feel shame **(Choice A)**.
- Initiative vs. guilt: Children (3–6 years) who successfully interact with others will develop a sense of initiative; those who are criticized will experience guilt.
- Industry vs. inferiority: Children (6–12 years) who successfully develop new skills will feel industrious; those who are not encouraged will feel inferior **(Choice B)**.
- Identity vs. role confusion: Adolescents (12–20 years) who successfully interact with peers develop a sense of self-identity; those who do not, experience role confusion.
- Intimacy vs. isolation: Adults (20–40 years) who can commit to and love others develop a sense of intimacy; those who do not, feel isolated **(Choice D)**.
- Generativity vs. stagnation: Adults (40–65 years) who successfully contribute to society feel productive; those who do not, feel stagnant.
- Integrity vs. despair: Older adults (>65 years) who feel accomplished gain a sense of integrity; those who do not, feel depressed and hopeless.

The older adult study participants, with a mean age of 78.2, should be dealing with the **integrity vs. despair** conflict associated with the final psychosocial stage of this theory.

Educational objective:
Erik Erikson's psychosocial theory of personality development includes eight stages that encompass the human life span. Each stage is associated with a crisis involving a conflict that provides an opportunity for growth. Successful resolution of stages results in healthy personality; unsuccessful resolution leads to psychopathology.

Time spent: 0 sec	**QID**:400380
Subject:	**System:**
Behavioral Sciences	Learning, Memory, and Cognition

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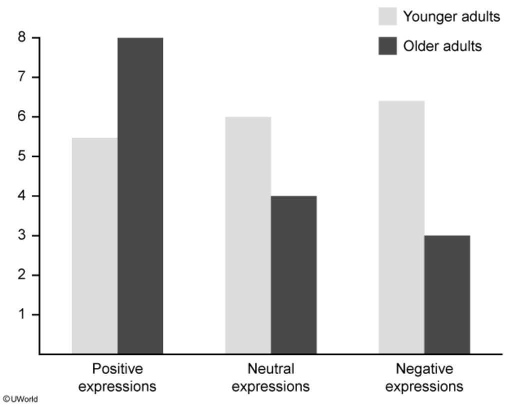

Figure 1 Average number of correctly identified faces for older and younger adult groups

Aging is associated with significant memory decline for all of the following types of memory EXCEPT:

- A. episodic. [14%]
- B. flashbulb. [28%]
- ✔ C. semantic. [52%]
- D. source. [5%]

Correct
52% answered correctly

Explanation:

Type of memory	Example memories	Effect of aging
Episodic	Buying first car, first day of college	Decline
Flashbulb	Where one was when the 9/11 terrorist attacks were announced	Decline
Source	What news source reported the story, who announced the information	Decline
Semantic	Names of people, colors, vocabulary	Stable
Procedural	Riding a bike, driving a car, serving a tennis ball	Stable

Aging affects different types of memory in various ways. **Aging** has been associated with **declines** in certain types of memory, including **episodic**, **flashbulb**, and **source memory (Choices A, B, and D).**

Episodic memory is the memory of autobiographical events, such as the name of the street one grew up on. Flashbulb memory is a type of episodic memory that is especially vivid and detailed. A flashbulb memory is a snapshot of the time and place when one received important historical or emotionally charged autobiographical news (eg, September 11, 2001). Source memory is remembering what source the information being remembered came from (eg, learning about an event from the newspaper or a friend).

Semantic and procedural types of memory appear to **remain relatively stable with age**. Semantic memory is the memory for words, facts, and concepts that have been acquired over the lifetime (eg, color names). Procedural memory involves motor skills that one has acquired (eg, riding a bike, driving a car).

Educational objective:

Aging has been associated with declines in certain types of memory, including episodic, flashbulb, and source memory. Semantic and procedural types of memory appear to remain relatively stable with age.

Time spent:0 sec
Subject:
Behavioral Sciences

QID:400381
System:
Learning, Memory, and Cognition

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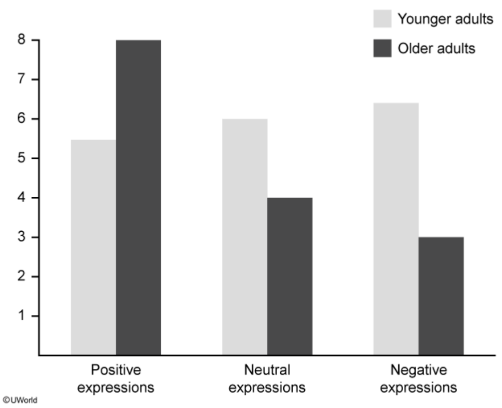


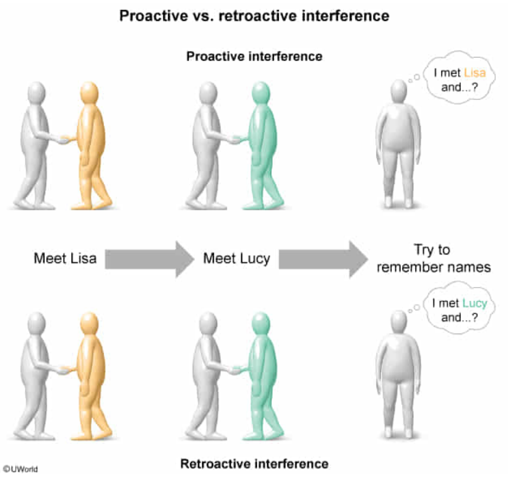
Figure 1 Average number of correctly identified faces for older and younger adult groups

Decreased memory caused by the distraction task demonstrates:

- A. anterograde amnesia. [2%]
- B. retrograde amnesia. [4%]
- C. proactive interference. [31%]
- D. retroactive interference. [60%]

Correct
60% answered correctly

Explanation:



Exposure to a distraction can interfere with the ability to accurately retrieve a memory or learn new information. This type of **interference** commonly contributes to nonpathological memory loss. **Retroactive interference** occurs when more **recent information interferes** with one's ability to recall older information. For example, a man is introduced to Lisa (first event) and then to Lucy (second event). Later, he cannot remember the name "Lisa" because "Lucy," the second name he heard, keeps interfering with his ability to remember the older information.

In the memory studies, the participants were asked to memorize neutral items (first event) and then subjected to a *distraction* (second event). The distraction, which was the more recent event, retroactively interfered with the participants' abilities to recall the items.

Proactive interference occurs when **previously learned information interferes** with one's ability to recall new information. For example, if one of the participants memorized a grocery list before the study (first event), and then had trouble recalling the items they were asked to memorize (second event) because they kept remembering items from their grocery list, this would be proactive interference **(Choice C)**.

(Choice A) Anterograde amnesia is a form of pathological memory loss in which a person cannot form new memories after the incident that caused the amnesia (eg, traumatic brain injury due to a car crash).

(Choice B) Retrograde amnesia is a form of pathological memory loss in which a person cannot access memories that were encoded before the incident that caused the amnesia (eg, a stroke).

Educational objective:

Retroactive interference occurs when recently encoded information prevents the recall of older information. Proactive interference occurs when previously encoded information prevents the recall of newer information.

Time spent:0 sec
Subject:
Behavioral Sciences

QID:400382
System:
Learning, Memory, and Cognition

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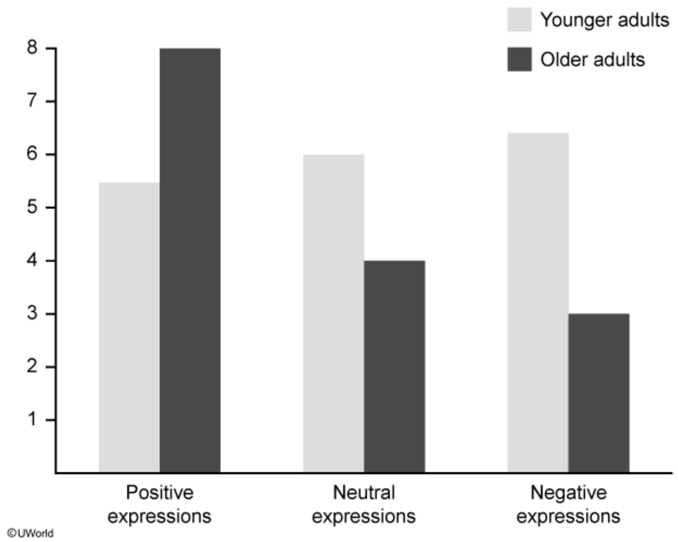

Figure 1 Average number of correctly identified faces for older and younger adult groups

If the researchers in the third study wanted to test the influence of aging on context-dependent memory effects, they could:

- A. show the faces to the older adult participants in one room, then move half of these participants to a new room to identify faces. [6%]
- B. show the faces to all participants in one room, then move half of the participants in each group to a new room to identify faces. [43%]
- C. provide background information about the faces to half of the older adult participants while faces are being memorized. [5%]
- D. provide background information about the faces to half of participants in each age group while faces are being memorized. [44%]

Correct
44% answered correctly

Explanation:

Context-dependent memory refers to the fact that individuals are better able to remember information when they are in the same context in which that information was learned. **Context** refers to the **physical environment** in which the original learning took place or the original memory was encoded. Context-dependent memory helps to explain why, when you meet people in one context (eg, in class) and then run into them in another context (eg, at Starbucks), you may have trouble remembering their names or how you know them.

In this example, if participants are first exposed to images of faces in one room, they are more likely to recognize those same faces if they are in the same room rather than if they are in a different room. This design must be performed on participants from both age groups so that this effect can be compared between the older and the younger participants. In this way, context-dependent memory effects in the younger adults will serve as a baseline to determine the influence of aging on context-dependent memory effects.

(Choice A) Breaking the older group into two groups would allow for the determination of context-dependent memory effects in the older group. However, if this design is not repeated in the younger group, there will be no baseline to determine how age influences context-dependent effects.

(Choices C and D) Background information or context provided about the pictures themselves would not influence context-dependent memory, which applies to the context in which memory is acquired and retrieved.

Educational objective:
Context-dependent memory is the process whereby information is more easily recalled when an individual is in the same context (eg, room, setting) where he or she first learned the information. To determine the influence of aging on context-dependent memory, a comparison between older and younger participants is needed.